METRIC
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COMMERCIAL ITEM DESCRIPTION

FIRE RETARDANT GEL, FOR SHIP CONSTRUCTION, MAINTENANCE, REPAIR, AND MODERNIZATION

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This Commercial Item Description (CID) covers fire retardant gel (FRG) for use during ship construction, maintenance, repair, and modernization including aboard ships. This CID does not apply to submarine construction, maintenance, repair, and modernization or in ship areas under the cognizance of NAVSEA 08. FRG is generally used as a temporary, protective layer to prevent thermal transfer or ignition of conventional organic coatings, fabrics, polymeric, or other combustible materials on surfaces surrounding hot work operations.

2. CLASSIFICATION. The FRG shall conform to the following types:

a. Type I – Pre-mixed/ready for use

b. Type II – Concentrate that must be mixed with potable water before use

3. SALIENT CHARACTERISTICS.

3.1 Description. FRGs (also known as water enhancers) are water-laden gels used for thermal protection. FRGs, either pre-mixed or mixed locally, can be applied to fabrics (e.g., blankets, curtains, and pads as defined by NFPA 51B) used to protect piping, surfaces, and equipment adjacent to hot work, such as welding, torch cutting, and grinding. The gel behaves like a thick blanket of water; it has a high heat capacity and high latent heat of evaporation, which enables it to withstand temperatures found during hot work. The gel effectively removes two elements from the fire triangle; it temporarily shields the fuel from both oxygen and heat.

3.2 Materials. When a definite FRG is not specified, the material selected shall be the best commercial grade for the purpose intended. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the FRG.

3.2.1 Unacceptable ingredients. FRGs used on Navy ships or in Naval shipyards shall not contain, or list on the Safety Data Sheet (SDS) for the product, any of the following materials:

a. Sodium ferrocyanide (yellow prussiate of soda [YPS])

b. Dichromate or any other compound containing hexavalent chromium

c. Thiourea

d. Borate or other boron-containing compounds, excluding barium sulfate (barites)

e. Polychlorinated biphenyl (PCB)

f. Polybrominated diphenyl ethers (PBDE)

g. Nonylphenol ethoxylates (NPE)

h. Per- and polyfluoroalkyl substances (PFAS)

Beneficial comments, recommendations, additions, deletions, clarifications, and any other data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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3.2.2 Detrimental materials. Although the FRG is likely to consist of environmentally acceptable materials (similar to the absorbent materials used in diapers), it will be approved and listed on the shipyard Authorized Use List (AUL) or equivalent database. The FRG shall have minimized concentrations of chemicals known to have a detrimental effect on welding processes, identified in the American Welding Society's (AWS) Welding Handbook or in shipyard-specific policies. These chemicals include the following:

a. Mercury
b. Antimony
c. Bismuth
d. Cadmium
e. Lead
f. Tin
g. Zinc
h. Bromides
i. Chlorides
j. Fluorides
k. Sulfur
l. Phosphorus

3.2.3 Physical and chemical properties. When specified (see 7.3), chemical or physical testing shall be performed. Lots or batches of FRG used on a Navy ship or in a Naval shipyard shall include an SDS that explains the physical and chemical properties of the material in accordance with appendix D of 29 CFR § 1910.1200, Hazard Communication.

3.2.4 Stability and reactivity. The FRG shall not have any reactivity hazards. The FRG shall be stable when exposed to the environment as specified in 3.2.9 below.

3.2.5 Toxicological information.

3.2.5.1 Toxicity. The manufacturer shall evaluate that the FRG does not pose known risk(s) to the health of personnel or the environment when used for its intended purpose. Additionally, the Navy will perform an evaluation assessment to confirm the manufacturer's testing results (see 3.2.5.2).

3.2.5.2 Health hazard assessment (HHA). An HHA will be conducted to ensure conformance to 3.2.5.1 as specified (see 7.3). The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the FRG using data provided by the manufacturer/distributor to the NMCPHC (see 3.2.5.1 and 7.4).

3.2.6 Ecological information. The FRG shall be inert in aerobic and anaerobic conditions, decompose or be immobile in landfills or soil systems, and have no chronic aquatic toxicity in the event of spills.

3.2.7 Disposal considerations. The FRG shall be able to be cleaned up with simple janitorial tools, including brooms, brushes, squeegees, vacuums, paper towels, wipes, absorbent cloths, and garbage bags. The FRG shall decompose or be disposable in waste landfills in concentrated form or in effluent water treatment plants in diluted form.

3.2.8 Shelf life. The FRG shall have a 1-year minimum shelf life both when undiluted and when diluted to the final use ratio. All packaging shall be clearly labeled with the expiration date.

3.2.9 Storage environment. The FRG shall comply with all requirements specified herein when subjected to extended storage temperatures between 10 and 55 °C (50 and 131 °F). Where the manufacturer's recommended minimum storage temperature is higher than 10 °C (50 °F) or the manufacturer's recommended maximum storage temperature is below 55 °C (131 °F), all packaging shall be clearly labeled with the manufacturer's recommended storage temperature range as shown on the commercial product data sheet.

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3.2.10 Corrosion. When applied to common shipbuilding materials, including those listed in the American Bureau of Shipbuilding's Generic Rules for Materials and Welding and Survey After Construction - Part 2, Rules for Materials and Welding, and NAVSEA S9074-AQ-GIB-010/248, Requirements for Welding and Brazing Procedure and Performance Qualification, the FRG (i.e., as mixed and ready to use) shall cause no more corrosion than would occur if the common shipbuilding materials were exposed to potable water for an 8-hour period.

3.2.11 Conductivity. The FRG, when mixed with potable water in accordance with the manufacturer's instructions, shall have a conductivity less than or equal to 50,000 micro-siemens per centimeter (µS/cm) per ASTM D4399.

3.2.12 Combustibility. The FRG shall be non-combustible in accordance with ASTM E136, method A. To meet the requirements of non-combustibility, the FRG shall be tested in a liquid holder and at least three of the four specimens tested shall exhibit no flaming from the test specimen at any time during the test.

3.3 Product safety. As specified by the procuring activity (see 7.3), chemical product safety will be communicated through an SDS in accordance with the GHS. The use of FRGs on Navy ships or in Naval shipyards shall not require any significant changes in safety practices or procedures. The FRG SDS will satisfy all minimum hazardous material reporting requirements in appendix D of 29 CFR § 1910.1200.

3.4 Hazards identification.

3.4.1 Health hazards. The FRG, when mixed with potable water in accordance with the manufacturer's instructions, and applied in accordance with the manufacturer's instructions, shall not cause any of the following health hazards as classified in appendix A of 29 CFR § 1910.1200:

a. Acute toxicity
b. Acute chemical burns, irritation, or sensitization of skin
c. Eye damage or irritation
d. Respiratory system damage or irritation
e. Germ cell mutagenicity
f. Reproductive toxicity
g. Carcinogen
h. Specific target organ toxicity
i. Aspiration hazard
j. Asphyxiation

3.4.2 Physical hazards. The FRG in final application form and the applied FRG shall not possess any of the following physical hazards as classified in appendix B of 29 CFR § 1910.1200:

a. Explosives
b. Flammable gases
c. Flammable aerosols
d. Oxidizing gases
e. Gases under pressure
f. Flammable liquids
g. Flammable solids
h. Self-reactive chemicals
i. Pyrophoric chemicals
j. Self-heating chemicals
k. Chemicals which, in contact with water, emit flammable gases
l. Oxidizing liquids
m. Oxidizing solids

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n. Organic peroxides

o. Corrosive to metals

p. Combustible dust

Additionally, the stored, unused FRG shall have the same requirements except the “gases under pressure” hazard restriction.

3.5 Hazards not otherwise classified. The FRG shall not have any hazards not otherwise classified as defined in 29 CFR § 1910.1200.

3.6 Design and operation.

3.6.1 Thermal performance. The FRG, when applied to a piece of cloth in accordance with MIL-C-24576, type I, class 2, per the manufacturer’s instructions, shall be capable of passing testing in accordance with FM 4950, Evaluating Welding Pads, Welding Blankets, and Welding Curtains for Hot Work Operations. The applied FRG on a piece of MIL-C-24576, type I, class 2, material shall meet the performance requirements of Welding Pads for the Fire and Thermal Resistant Test and the Paper Ignition Test in accordance with FM 4950. However, the accelerated weathering test and charring embrittlement testing requirements of FM 4950 shall not apply. The FRG shall be tested twice. The first test series shall be completed at time 0 (immediately after application onto required cloth). A second test series shall be completed when the maximum effective service period of the FRG has been reached (e.g., 60 minutes, 120 minutes, etc.). The accelerated weathering test and charring embrittlement testing requirements of FM 4950 shall not apply.

3.6.2 Product dilution (if applicable). Where concentrated product is intended to be diluted or mixed with potable water for distribution into smaller containers by the end user, the packaging shall be clearly labeled with the manufacturer’s instructions for dilution. Diluted material shall have a minimum shelf life the same as the concentrated product (see 3.2.8); where dilution is capable of extending the shelf life of the concentrated product, the packaging shall be labeled with instructions for how to determine the revised expiration date based on the concentrated product expiration date (see 3.2.8).

3.6.3 Application methods and tools. Packaging shall be labeled with the manufacturer’s instructions for coverage rates and for applying the FRG. Application tools should be limited to a spray bottle or other portable spray equipment, paint or other brushes, or paint rollers with a 9-millimeter (⅜-inch) or greater nap. The FRG shall be formulated to allow removal from application tools by manual cleaning of the tools with detergent and potable water.

3.6.4 Effective service period. The applied FRG shall remain effective for at least 60 minutes following application in accordance with the manufacturer’s recommended application instructions when applied to a surface in any orientation in an environment with an ambient relative humidity of 45 to 55 percent and an ambient temperature of 18 to 24 °C (65 to 75 °F).

3.7 Chemical contaminants and byproducts regarding welding. The FRG shall have minimized concentrations of chemicals known to have a detrimental effect on welding processes; identified in the AWS’s Welding Handbook or in shipyard specific policies (see 3.2.1 and 3.2.2). The user of the FRG shall not be exposed to hazardous chemicals in excess of the permissible exposure limits established in 29 CFR § 1910.1200 Subpart Z, Toxic and Hazardous Substances.

3.8 Removal.

3.8.1 Removal from fabrics. The FRG shall be able to be removed from personal protective equipment, industrial fabrics, and personal clothing via either normal household and industrial laundry process or industrial dry cleaning process.

3.8.2 Removal from smooth surfaces. The FRG shall be able to be removed with simple janitorial tools, including brooms, brushes, squeegees, vacuums, paper towels, wipes, absorbent cloths, and garbage bags as required by 3.2.7. Additionally, the FRG shall be able to be removed with methods that match the Society for Protective Coating’s (SSPC) SP 1, Solvent Cleaning; SSPC-SP 2, Hand Tool Cleaning; SSPC-SP 3, Power Tool Cleaning; SSPC-SP 6/NACE No. 3, Commercial Blast Cleaning; and SSPC-SP 7/NACE NO. 4, Brush-Off Wet Blast Cleaning.

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3.8.3 Removal from rough surfaces. The FRG shall be able to be removed from rough surfaces with a flushing of potable water. An additional removal chemical, such as detergent or solvent, may be used to assist removal from rough surfaces.

4. REGULATORY REQUIREMENTS. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

5.2 Market acceptability. The item offered shall have been sold to the Government or commercial market. A modified commercial, off-the-shelf item may be used to minimize design and engineering risk. The offeror shall certify and provide supporting documentation as to the product's serviceability.

5.3 Bid samples. When specified (see 7.3), a bid sample shall be submitted to the Government for examination and testing before the contract is awarded. The bid sample shall be non-returnable. Testing will be performed on the bid sample as applicable. Failure of the bid sample to meet the requirements of any of the applicable tests or examinations listed herein shall be cause for rejection of the bid. If required by the procuring activity, the government may request a complete data package showing that the FRG satisfies all the requirements cited in this CID.

5.4 Responsibility for inspection. Unless otherwise specified (see 7.3), the contractor is responsible for the performance of all inspection, examination, and test requirements specified herein. Unless otherwise specified (see 7.3), the contractor may use his own or any other facilities suitable for the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections, examinations, or tests set forth in this CID where such inspections, examinations, and tests are deemed necessary to ensure supplies and services conform to prescribed requirements. Failure of any of the tests and examinations listed herein shall be cause for rejection.

5.5 Testing and examination. The proposer is responsible for conducting all tests as specified herein. An independent testing laboratory that is accredited to ISO/IEC 17025 or equivalent procedure, and certified by an American Bureau of Shipping (ABS) surveyor or the Technical Authority (see 7.7), shall conduct all tests specified in this document. Accreditation to ISO/IEC 17025 shall be obtained from a recognized accreditation body, such as the American Association for Laboratory Accreditation (www.a2la.org) or International Code Council's International Accreditation Services (www.iasonline.org). The scope of accreditation shall include fire tests and flammability required for product validation. All other fire test provisions shall be as specified (see 7.3).

5.6 Material effects and corrosion. When specified (see 7.3), an FRG product delivered as a gel concentrate shall be tested for corrosion as required by 3.2.10 and 3.7. In addition, the gel concentrate shall be diluted with potable water in accordance with the manufacturer's instructions to prepare the final FRG that will be applied to the work areas, and that material shall also be tested for uniform corrosion as required by 3.2.10 and 3.7. Metallic materials used by the Navy are identified in 3.2.10.

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6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order. If the product is delivered in the final application container (e.g., squirt bottle), the containers shall arrive in boxes suitable for direct on the shelf storage and be suitable for palletization at a minimum. If the product is not delivered in its final application form, it shall be packaged as bulk material. The exterior of the package shall be labeled with the chemical contents, and the required safety markings in accordance with 29 CFR § 1910.1200.

7. NOTES.

7.1 Intended use. The FRG covered by this CID is intended to reduce the risk of shipboard fires. The FRG is intended to increase the level of fire protection to areas adjacent to hot work sites. It is intended for use by shipyard and ship's force personnel during ship maintenance activities where permissible. The FRG is not intended to be used for extinguishing fires and does not replace portable fire extinguishers; additionally, the FRG does not alleviate any fire prevention requirements of 29 CFR § 1915, Subpart P, Fire Protection in Shipyard Employment, or the requirements of any other invoked standard, specification, or contract item applicable to the subject maintenance, repair, or construction effort. When acquiring the FRG, the contracting activity will ensure that application requirements are specified (see 7.3) and that product is only used as directed.

7.2 Source of documents.

7.2.1 American Bureau of Shipping (ABS). ABS standards are available from www.eagle.org or the American Bureau of Shipping, Government Operations Office, 1421 Prince Street, Suite 100, Alexandria, VA 22314.

7.2.2 American Welding Society (AWS). AWS standards are available from www.aws.org or the American Welding Society, 550 N.W. LeJeune Road, Miami, FL 33126.

7.2.3 ASTM International. ASTM standards are available from www.astm.org or ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959.

7.2.4 Code of Federal Regulations (CFR). The CFR is available from www.ecfr.gov or the Superintendent of Documents, U.S. Government Publishing Office, Washington, DC 20401.

7.2.5 Defense specifications. Defense specifications are available from https://quicksearch.dla.mil.

7.2.6 FM Approvals. FM 4950 is available from www.fmapprovals.com.

7.2.7 International Organization for Standardization (ISO). ISO standards are available from www.iso.org or the International Organization for Standardization, Chemin de Blandonnet 8, CP 401, 1214 Vernier, Geneva, Switzerland.

7.2.8 National Fire Protection Association (NFPA). NFPA standards are available online at www.nfpa.org or the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02169.

7.2.9 Naval Sea Systems Command (NAVSEA) Publications. NAVSEA Publications are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. NAVSEA Publications are also available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.

7.2.10 Society for Protective Coatings (SSPC). SSPC standards are available from www.sspc.org or SSPC Publication Sales, 800 Trumbull Drive, Pittsburgh, PA 15205.

7.2.11 United Nations Economic Commission for Europe (UNECE). The Globally Harmonized System of Classification and Labeling (GHS) is available from https://unece.org/ghs-rev2-2007.

7.3 Ordering data. The contract or order should specify the following:

a. CID document number and revision.

b. Type of FRG (see section 2).

c. When chemical testing is required (see 3.2.3).

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d. When physical testing is required (see 3.2.3 and 5.6).
e. Requirements for an HHA (see 3.2.5.2).
f. Product conformance provisions.
g. Requirements for a SDS (see 3.3 and 6).
h. When bid samples are required (see 5.3 and 7.4).
i. Responsibility of inspection requirements if other than specified herein (see 5.4).
j. Additional fire testing provisions (see 5.5 and 7.5).
k. Packaging requirements.
l. Application requirements (see 7.1).

7.4 Toxicity evaluation. The NMCPHC requires sufficient information to perform an HHA of the FRG product. The HHA process is described on the NMCPHC's website, https://www.med.navy.mil/Navy-Marine-Corps-Public-Health-Center/.

7.5 Additional fire testing provisions. The Technical Authority reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed salient characteristics. A2LA (www.A2LA.org) or International Code Council's IAS (www.IASonline.org) maintain a full list of the approved laboratories that are accredited to ISO/IEC 17025 and provide details on how to obtain and maintain accreditation (see 7.3).

7.6 SDS. When specified (see 7.3), contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with appendix D of 29 CFR 1910.1200. In order to obtain the SDS, federal acquisition regulation clause 52.223-3 will be in the contract. The contracting activity should be given an SDS at the time of contract award.

7.7 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

7.8 Key words.

Fire protection
Heat protection
Hot work protection
Weld safety

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CONCLUDING MATERIAL

MILITARY INTERESTS:

Custodians:
Army – AV
Navy – SH
Air Force – 03
DLA – CQ

Review Activities:
Army – MI
Navy – AS
DLA – CC

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FAS

Preparing Activity:
Navy – SH
(Project 4210-2021-003)

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